جامعة حفر الباطن
University of Hafr Al Batin

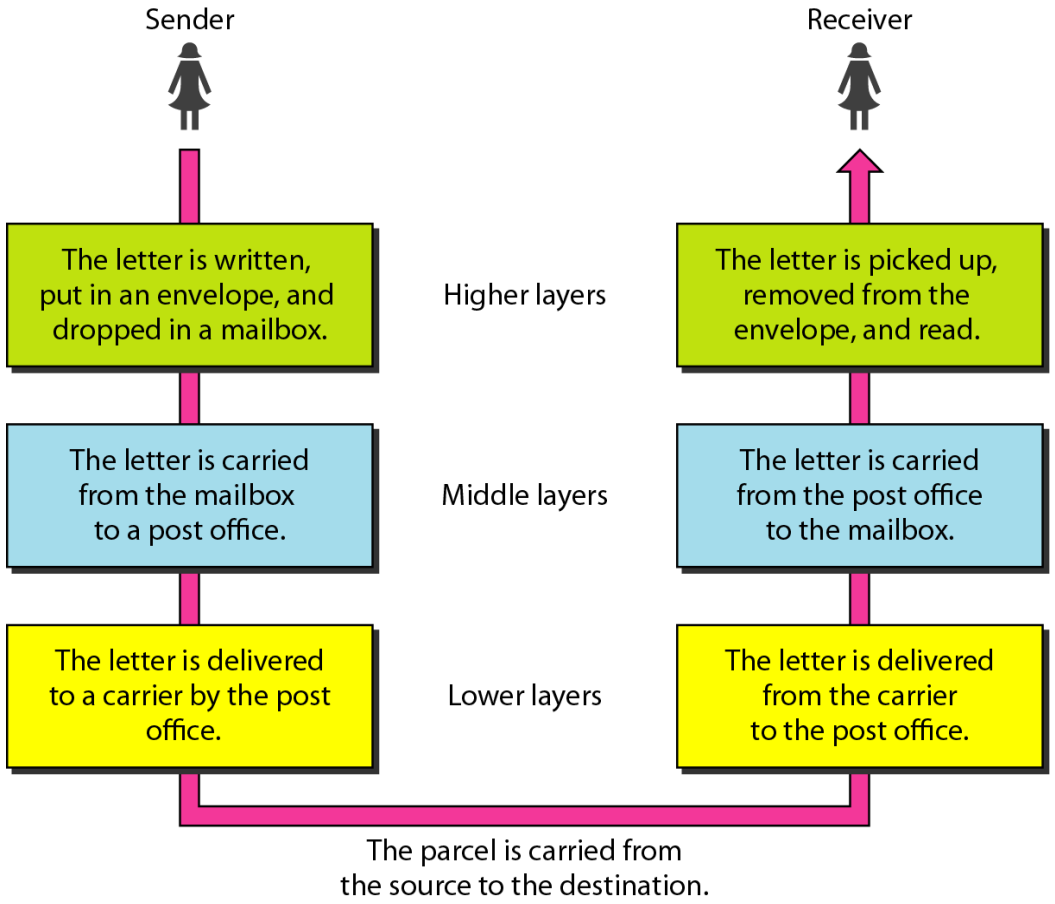
# Lecture 3 - Introduction to Network Models

# Objectives

- The Layering Concept
- OSI Model
- TCP/IP Protocol Suite
- Protocols in TCP/IP Layers
- Addressing in TCP/IP

# Why we use Layers?

- We use the concept of layers in our daily life. As an example, let us consider two friends who communicate through postal mail.
- The process of sending a letter to a friend would be complex if there were no services available from the post office.



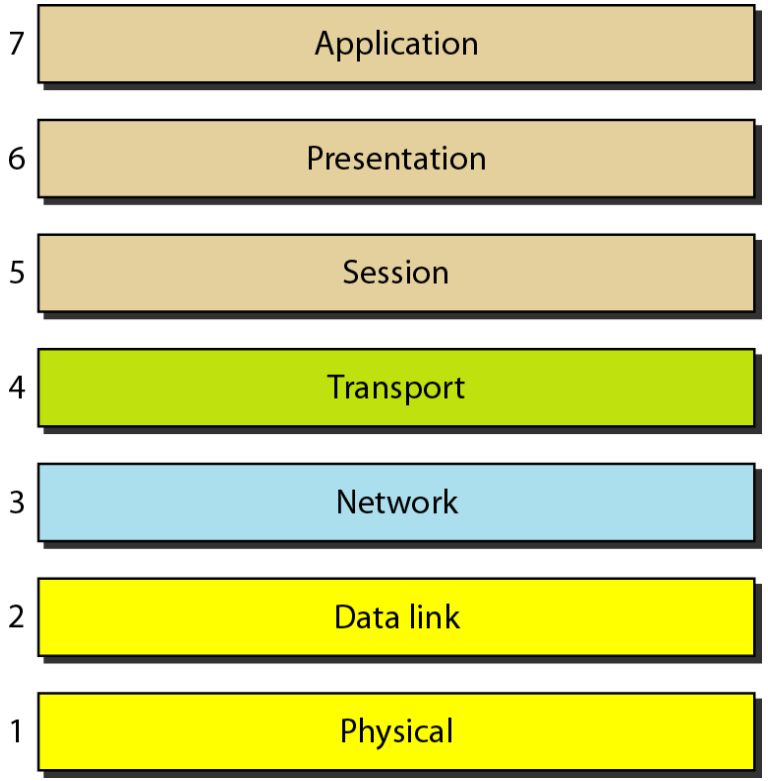
# OSI Model

# OSI Model

- Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards.
- An ISO standard that covers all aspects of network communications is the **O**pen **S**ystems **I**nterconnection (OSI) model.
  - It was first introduced in the late 1970s.
- **ISO is the organization, OSI is the model.**

# OSI Model

- OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems
  - It is not a protocol
- It consists of **seven separate but inter related layers** each of which define a part of the process of moving information across a network



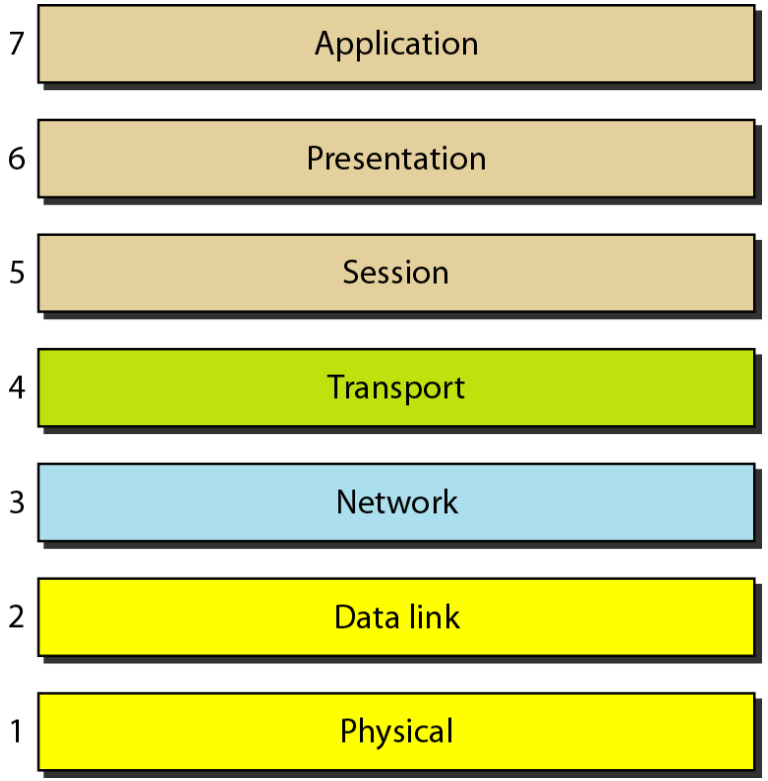
# Layers in OSI Model

## 1. Physical Layer:

Responsible for movements of individual **bits** from one hop (node) to the next.

## 2. Data link Layer:

Responsible for moving **frames** from one hop (node) to the next.



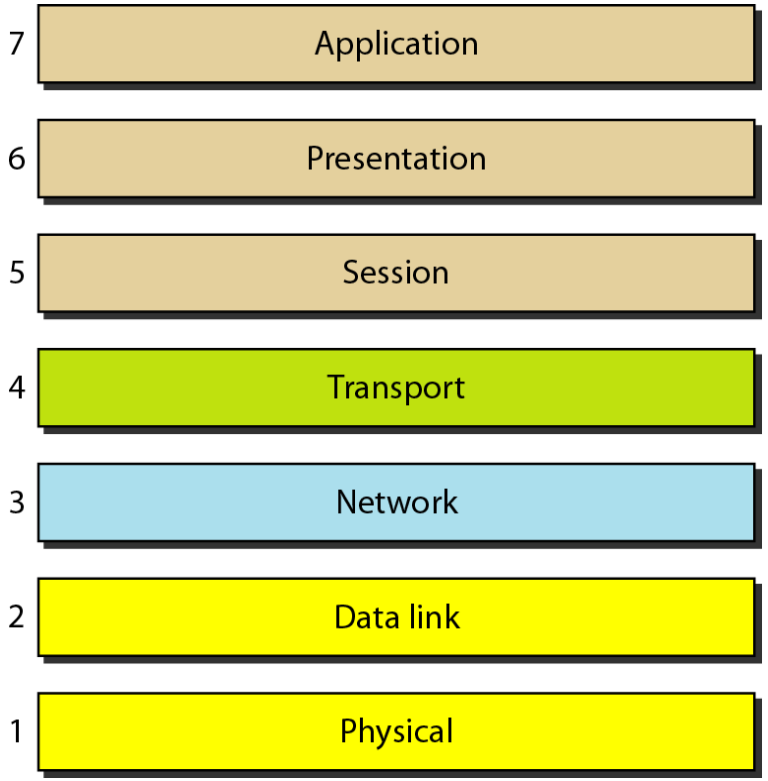
# Layers in OSI Model

## 3. Network Layer:

Responsible for the delivery of individual **packets** from the source host to the destination host.

## 4. Transport Layer:

Responsible for the delivery of a **message** from one process to another.



# Layers in OSI Model

**5. Session Layer:**

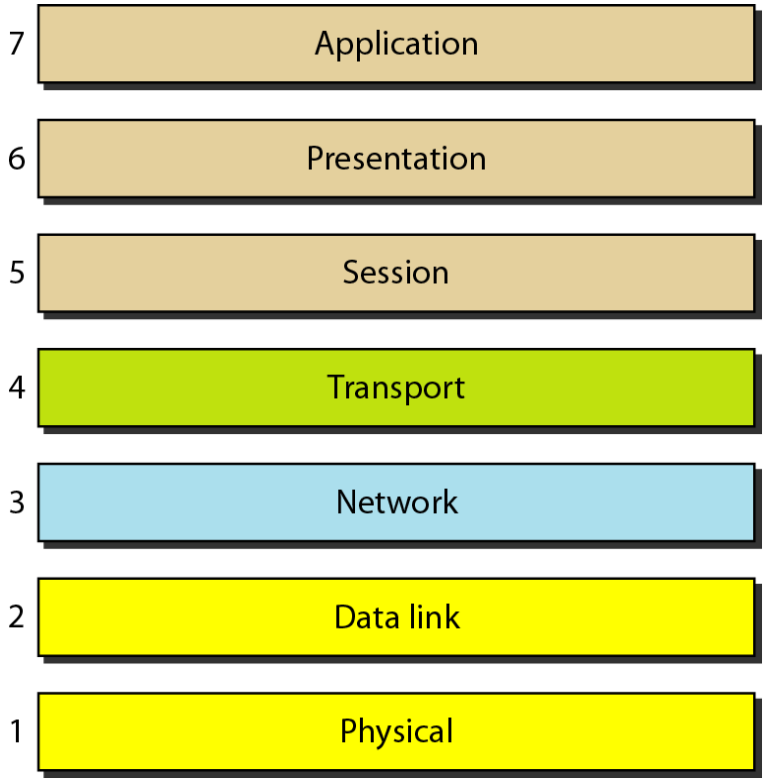
Responsible for establishing, managing, and controlling sessions.

**6. Presentation Layer:**

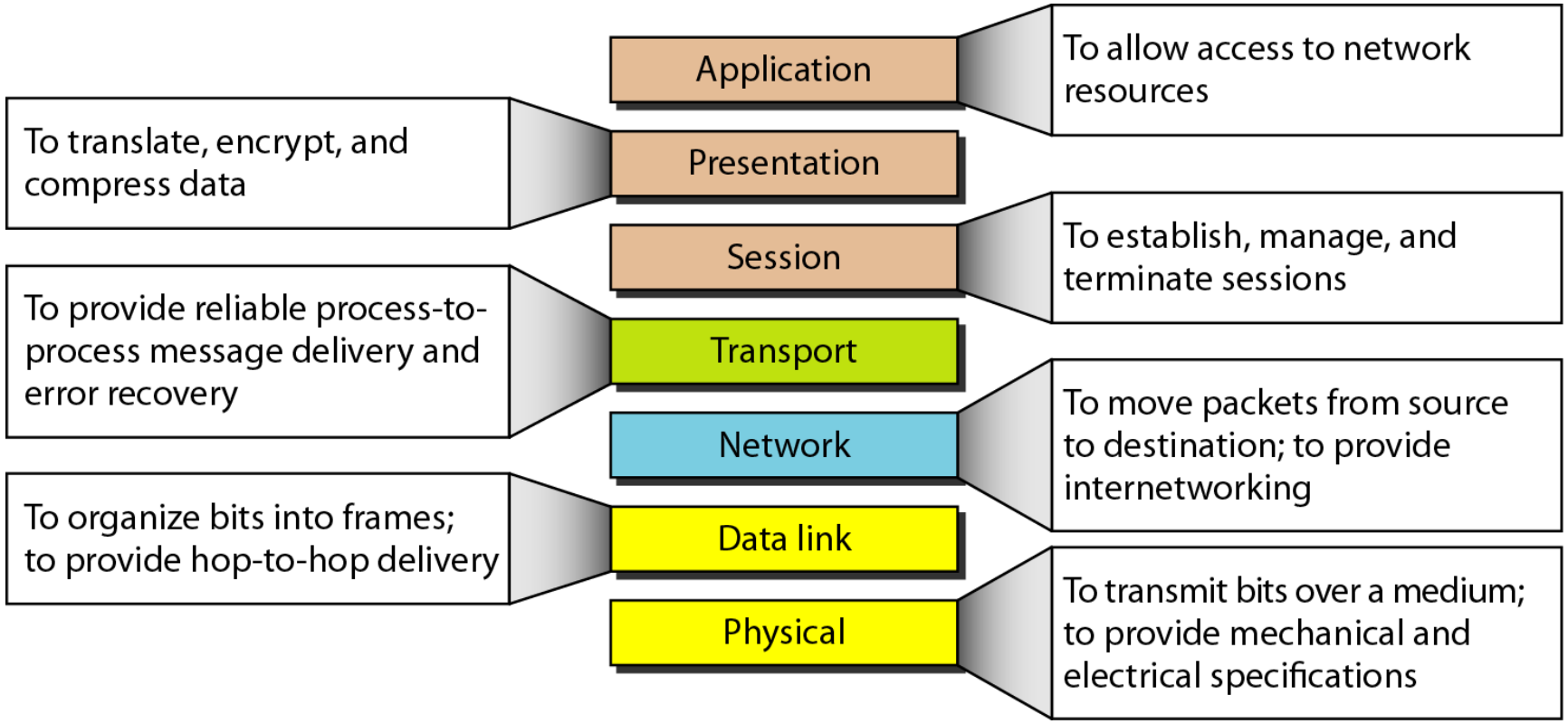
Responsible for translation, compression, and encryption.

**7. Application Layer:**

Responsible for allowing access to network resources and providing services to the user.



# Layers in OSI Model



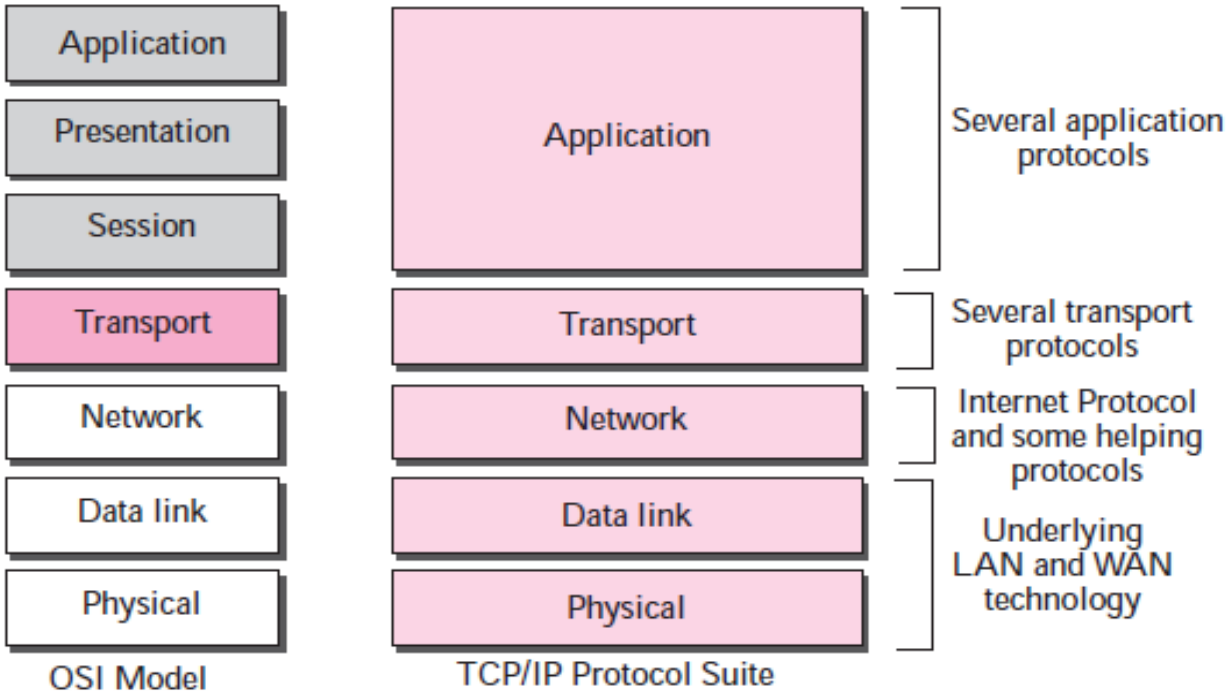
# TCP/IP Protocol Suite

# TCP/IP Protocol Suite

- The layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- The original TCP/IP protocol suite was defined as having four layers:
  1. Host-to-network
  2. Internet
  3. Transport
  4. Application

# TCP/IP Protocol Suite

- However, when TCP/IP is compared to OSI, we can say that the **TCP/IP protocol suite is made of five layers**:
  1. Physical
  2. Data link
  3. Network
  4. Transport
  5. Application.



# Protocols in TCP/IP Layers

## Protocols in The Network Layer:

- The main protocol here is the **Internet Protocol (IP).**
- There are some **auxiliary protocols** that helps the IP in the delivery and routing task:
  1. **Internet Control Message Protocol(ICMP):** helps IP to report any problem during packet routing
  2. **Dynamic Host Configuration Protocol (DHCP):** helps IP to get the network layer address for the host.
  3. **Address Resolution Protocol (ARP):** helps IP to find the link layer address of a host or a router when it network layer address is given.

# Protocols in TCP/IP Layers

## Protocols in The Transport Layer:

1. The main protocol here is **Transmission Control Protocol (TCP)**: a connection-oriented protocol that first establishes a logical connection at the two hosts transport layer before transferring data
2. **User Datagram Protocol (UDP)** is a another simple protocol in the Transport Layer.
   - It does not provide flow control, error control and congestion control
3. **Stream Control Transmission Protocol (SCTP)** is a new protocol designed to respond to new application in multimedia

# Protocols in TCP/IP Layers

## Protocols in The Application Layer:

- The application layer in the Internet includes many predefined protocol

1. **Hypertext Transfer Protocol (HTTP):** used for accessing World Wide Web (WWW)
2. **Simple Mail Transfer Protocol (SMTP):** the main protocol for electronic email
3. **File Transfer Protocol (FTP):** used for transferring files from one host to the other
4. **Terminal Network (TELNET)** and **Secure Shell (SSH):** used for accessing site remotely

# Protocols in TCP/IP Layers

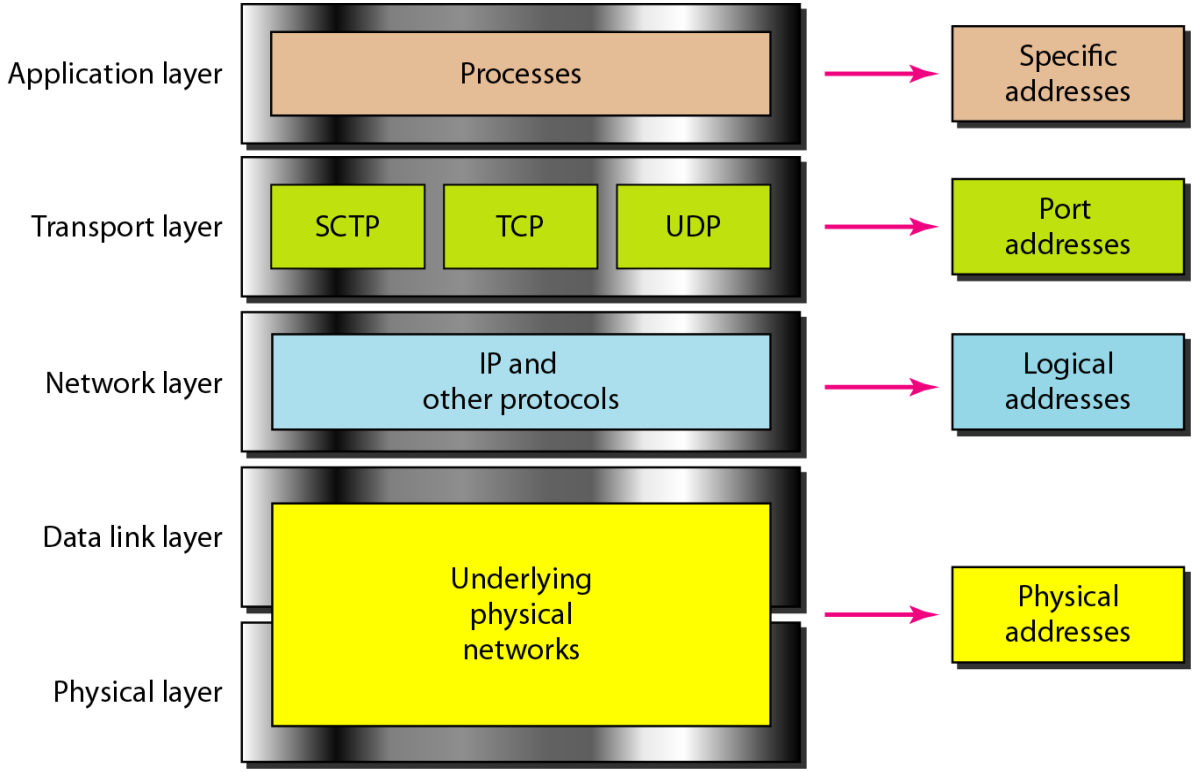
## Protocols in The Application Layer: (cont.)

5. **Simple Network Management Protocol (SNMP):** used by an administrator to manage Internet at global and local level
6. **Domain Name System (DNS):** used by other protocols the network layer address of a computer
7. **Internet Group Management Protocol (IGMP):** used to collect membership in a group

# Addressing in TCP/IP

# Addressing in TCP/IP

- Four levels of addresses are used in an internet employing the TCP/IP protocols:
  1. Physical Addresses
  2. Logical Addresses
  3. Port Addresses
  4. Specific Addresses
- Each address is related to a particular layer



# Addressing in TCP/IP

## 1. Physical Address:

- Physical address is the address of a node defined by its LAN or WAN.
  - Also called Link Address or MAC Addresses.
  - The lowest level address
  - It is included in the frame used by the data link layer

- The size and the format depends on the network.
  - For example, Ethernet uses a 6-byte (48 bit) physical address imprinted on the Network Interface Card (NIC)

**07:01:02:01:2C:4B**

A 6-byte (12 hexadecimal digits) physical address.

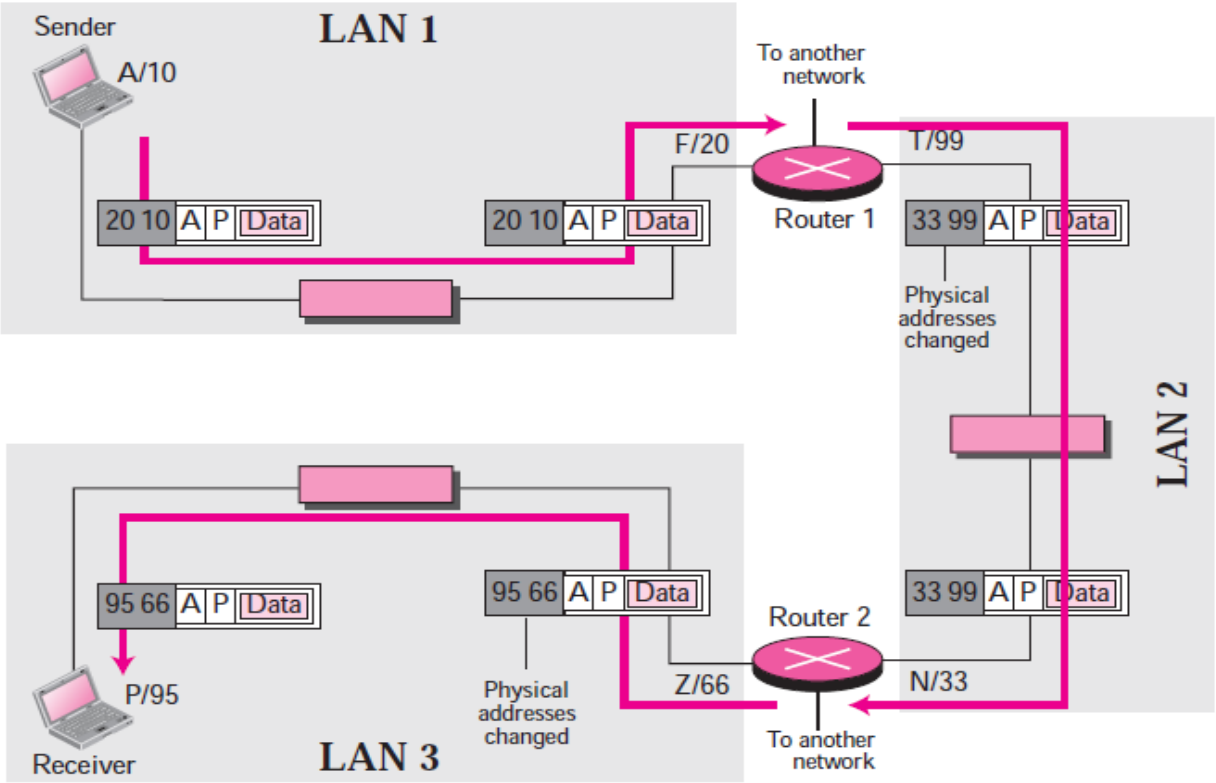
# Addressing in TCP/IP

## 2. Logical Address:

- A logical address in the Internet is the current 32-bit address that uniquely identify every host in the Internet
  - Also called the IP address.
- The physical address will change from hop to hop but the logical address remains the same.
- The logical address can be unicast (single recipient), multicast (multiple recipient) and broadcast (all host in the network).
- logical address used in internet is currently a 32-bit address (IPv4)

# Addressing in TCP/IP

## 2. Logical Address:



# Addressing in TCP/IP

## 3. Port Address:

- A port address is a 16-bit address represented by one decimal number.
  - The physical addresses change for each trip a packet takes, but the logical and port addresses usually remain the same.
  - The port number can range from 0 to 65535
- The IP and physical address are necessary for data travel from one computer to another
- But, arrival at the destination is not the final objective
  - The main objective of internet is the process to process communications. For this purpose it is necessary to label or name the process.
- **Thus the process need addresses!!**

**753**

A 16-bit port address represented as one single number.

# Addressing in TCP/IP

## 4. Application Specific Address:

- Some applications have user friendly address that are designed for specific applications.
- Examples:
  - Email address (turki@uhb.edu.sa)
  - Universal Resource Locator (URL) – www.google.com